

WINNIPEG INTL, MB, Canada

WMO: 718520

Lat: 49.9158N

Lon: 97.2493W

Elev: 783

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.5	(c) -21.2	(d) -33.9	(e) 0.8	(f) -25.2	(g) -29.6	(h) 1.1	(i) -20.9	(j) 32.1	(k) 8.3	(l) 29.5	(m) 9.5	(n) 7.4	(o) 290	(p) 0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.1	(c) 86.6	(d) 70.2	(e) 83.6	(f) 68.5	(g) 80.9	(h) 66.8	(i) 73.6	(j) 82.9	(k) 71.1	(l) 80.4	(m) 69.0	(n) 77.7	(o) 13.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	115.3	79.9	67.9	105.4	76.7	65.7	97.7	74.2	37.8	83.1	35.6	80.5	33.7	77.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.9	22.2	-31.8	92.8	5.6	3.6	-35.8	95.4	-39.1	97.5	-42.2	99.5	-46.2	102.1
			-32.0	77.7	5.5	2.4	-36.0	79.4	-39.2	80.8	-42.3	82.1	-46.3	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	2.7	6.7	21.3	39.0	52.0	63.0	67.5	65.8	56.7	42.1	24.8	9.3
	DBStd	25.45	14.88	14.02	14.37	11.33	9.31	7.13	5.89	6.41	8.50	8.97	12.03	14.03
	HDD50	6345	1466	1213	891	356	90	3	0	0	32	277	757	1260
	HDD65	10254	1931	1633	1354	780	413	118	43	69	271	710	1207	1725
	CDD50	1870	0	0	2	26	151	392	542	491	233	33	0	0
	CDD65	303	0	0	0	0	10	57	119	94	22	1	0	0
	CDH74	3039	0	0	0	8	167	563	1105	963	221	12	0	0
	CDH80	788	0	0	0	0	39	137	284	281	45	2	0	0
	WSAvg	10.9	11.1	11.1	11.5	11.7	11.7	10.3	9.3	9.7	10.7	11.6	11.5	11.0
(15) Precipitation	PrecAvg	20.6	0.7	0.6	1.0	1.4	2.4	3.3	3.0	3.1	2.0	1.4	0.9	0.8
	PrecMax	30.6	2.0	1.9	2.7	3.9	7.1	9.1	7.6	8.7	5.9	5.1	2.4	1.7
	PrecMin	9.3	0.1	0.0	0.0	0.0	0.3	0.2	0.8	0.2	0.4	0.1	0.0	0.1
	PrecStd	4.4	0.5	0.5	0.6	1.0	1.8	1.9	1.8	2.0	1.2	1.1	0.6	0.4
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	39.6	60.0	74.4	85.3	89.2	90.0	90.9	85.4	74.5	56.2	38.8
		MCWB	34.0	37.0	50.5	54.8	62.4	72.2	73.6	71.3	67.5	60.3	48.1	34.8
	2%	DB	33.2	35.5	49.3	68.0	78.6	83.6	86.0	86.4	80.0	66.9	49.5	34.4
		MCWB	31.5	33.8	43.5	51.8	58.8	67.0	71.5	70.0	65.1	55.8	43.7	32.2
	5%	DB	28.9	32.7	43.3	63.1	73.5	80.1	83.1	82.9	75.7	61.3	44.8	31.5
		MCWB	27.4	31.4	39.2	49.5	57.2	65.6	69.8	68.3	63.3	52.5	40.4	29.9
	10%	DB	24.5	27.7	38.5	57.4	68.6	76.8	80.4	79.6	71.2	56.3	40.0	28.5
		MCWB	23.2	26.0	35.6	46.6	54.8	63.9	68.2	66.3	60.9	49.8	36.7	26.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	36.9	52.3	57.3	66.2	75.5	77.2	75.7	71.6	62.0	49.3	35.4
		MCDB	36.8	39.3	59.3	70.9	76.2	85.6	86.8	85.1	80.7	72.7	54.0	38.0
	2%	WB	31.5	33.9	43.8	52.9	61.9	70.7	74.3	72.3	67.6	57.4	44.6	32.5
		MCDB	33.3	35.3	49.1	65.4	71.8	79.3	82.8	82.7	75.5	64.8	48.7	34.2
	5%	WB	27.5	31.2	39.4	50.1	59.4	67.8	72.0	70.0	64.8	53.6	40.5	29.8
		MCDB	29.0	32.9	43.1	60.5	69.0	76.6	80.9	79.9	72.7	59.6	44.2	31.4
	10%	WB	23.2	26.1	35.9	47.3	57.0	65.4	69.5	67.9	62.1	50.0	36.8	27.0
		MCDB	24.5	27.6	38.3	57.0	66.5	73.8	77.8	77.0	69.8	55.8	39.8	28.4
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	17.3	17.2	20.7	22.5	20.3	21.1	22.3	20.9	17.1	14.5	14.8
		MCDBR	16.4	14.6	19.2	29.0	29.3	24.9	23.6	26.2	26.9	25.1	20.4	16.0
		MCWBR	15.4	13.2	14.1	16.8	15.1	12.7	12.7	13.0	14.8	15.5	14.9	14.4
	5% WB	MCDBR	16.2	14.2	18.4	26.1	23.1	20.7	20.9	22.7	22.5	22.7	18.7	15.5
		MCWBR	15.3	12.9	13.9	16.0	13.7	12.2	12.6	12.6	13.9	14.9	14.1	14.1
(40) Clear-Sky Solar Irradiance	taub		0.265	0.301	0.309	0.356	0.373	0.385	0.392	0.380	0.347	0.306	0.274	0.250
	taud		2.253	2.140	2.293	2.333	2.352	2.342	2.338	2.368	2.455	2.536	2.463	2.333
	Ebn at Noon		247	261	285	282	281	277	273	271	270	265	248	239
	Edh at Noon		22	32	34	36	37	38	38	35	29	22	18	18
(44) All-Sky Solar Radiation	RadAvg		428	814	1216	1590	1729	1850	1949	1606	1157	683	377	303
	RadStd		46	59	100	119	143	144	81	107	90	93	53	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	-1.01	-1.44	N/A	N/A	N/A	N/A

Nomenclature: See separate page